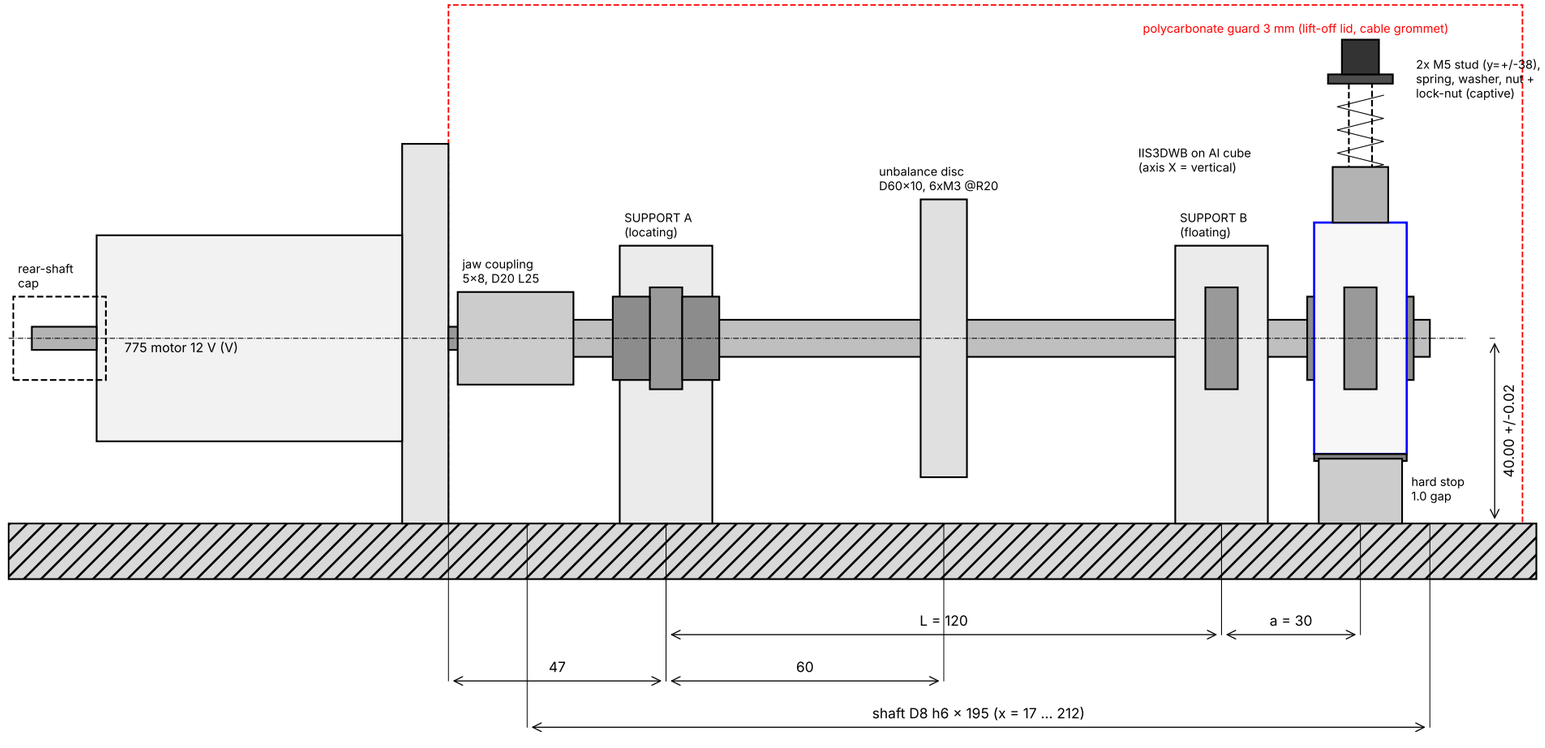
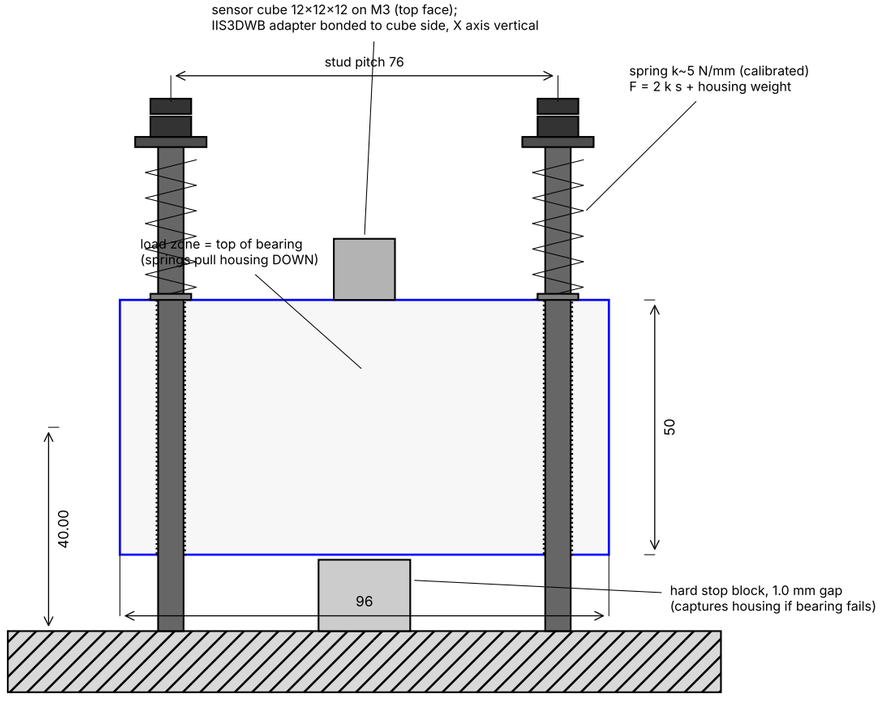
VIEW A - side elevation (x-z), guard shown dashed

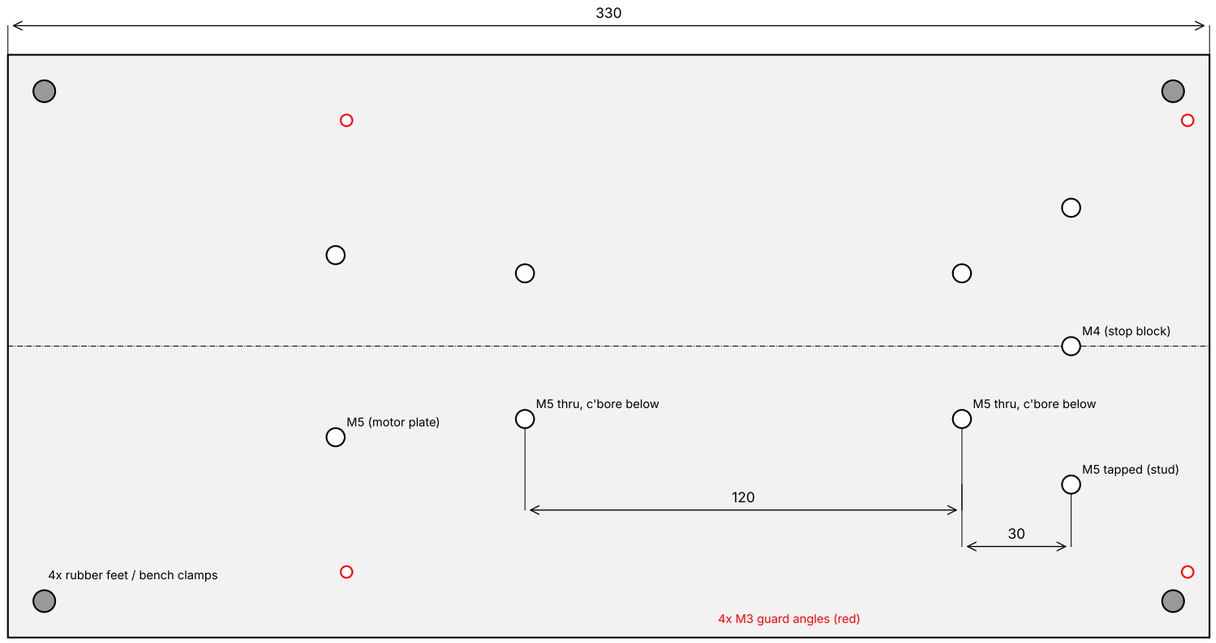


VIEW B - test housing and captive load (y-z), looking from shaft end



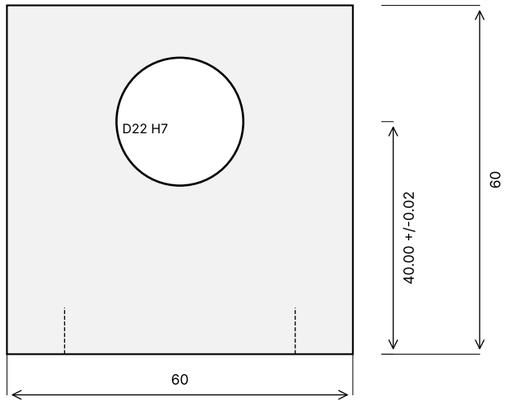
Load set by measuring washer height above housing top (vernier) on both studs; equal within 0.1 mm.
Studs M5 in tapped base holes + lock-nut below; holes in housing D5.6 (allows vertical travel and 41' tilt).

VIEW C - base plate (x-y), 330 × 160 × 12 Al 6061, holes



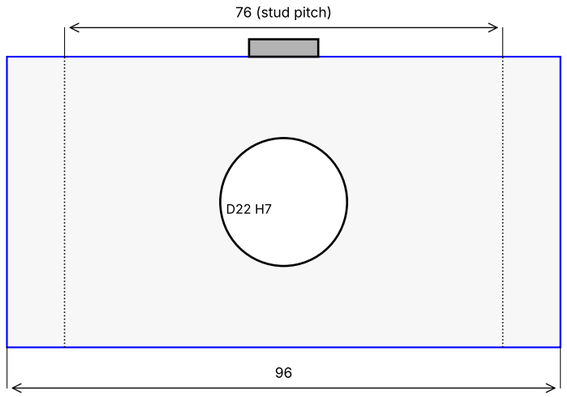
IVHM bearing rig - Design B - assembly
Rev A, 22 Oct 2026 - sheet 1/2 - all dims mm
General tol. ISO 2768-m; (V) = verify on purchased part before machining

P1 SUPPORT HOUSING x2 (Al 6061-T6, 60 × 20 × 60)



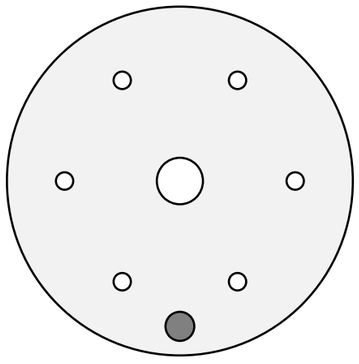
Bore D22 H7 (+0.021/0) through, Ra 0.8
Bore axis to base face: 40.00 +/-0.02 (all three housings line-bored in ONE setup, clamped together)
Base face flat 0.01; bore perpendicular to faces 0.02
2x M5 tapped 10 deep in base face at y = +/-20
4x M3 tapped 6 deep each face on PCD 30 for cover plates (P4)
Support A: covers both faces (locating). Support B: cover on one face only (axial float)

P2 TEST HOUSING x1 (Al 6061-T6, 96 × 20 × 50, floating)



Bore D22 H7 through, centred (25.00 from bottom face), line-bored with P1 (shim 15.00 under it)
2x D5.6 through (vertical) at y = +/-38 for M5 studs; PTFE washer under each spring
M3 × 6 deep tapped at top-face centre (sensor cube); top face flat 0.02
4x M3 for cover plates each face (cartridge: housing + 608 + 2 covers)
Mass ~ 0.24 kg (adds 2.3 N to the spring load)

P3 UNBALANCE DISC (Al, D60 × 10)



Bore D8 H7; 2x M4 set screws at 90 deg (cup point, thread-locked)
6x M3 tapped through on R20 at 60 deg (unbalance screws M3×6, ~0.5 g; M3 nut adds ~0.3 g)
Magnet pocket D5 × 2 deep at R25 (tach); balance check: run with no masses, record 1x level
Faces parallel 0.02; run-out after mounting <= 0.03 TIR (dial indicator)

P4 COVER PLATES x6 (Al 2 mm, D34)

Centre hole D20 (clears collars D18); 4x D3.4 on PCD 30
Clamp outer ring axially; no contact with inner ring

P5 MOTOR PLATE (Al 10 thick, 80 × 70, + gusset)

Pilot bore for 775 front boss: D (V) H8, centre 40.00 +/-0.02 above base face
2x motor screw clearance holes on the 775 face pattern (V) (typ. M4 on 29 mm)
2x M5 tapped into bottom edge + 1 gusset; plate face perpendicular to base 0.05
Shim pack 0.05/0.1 under plate for final centre-height match

P6 STOP BLOCK (Al 18 × 20 × 14), P7 SENSOR CUBE (Al 12 × 12 × 12, M3 clearance)

Stop block top 14.00 → 1.0 mm gap to housing (15.00); cube faces square 0.05

BOUGHT-OUT / ASSEMBLY NOTES

Shaft: D8 h6 induction-hardened linear rod, cut 195, ends chamfered 0.5×45; micrometer check 7.991-8.000
Bearings: SKF 608-2Z, one batch; mark serials; press only on the ring being fitted (sleeve tools)
Shaft collars D8 clamp type x4 (A both sides, test inboard + outboard); torque recorded
Coupling jaw 5×8 D20 L25; set screws on motor flat / shaft, thread-locked
Alignment: dial indicator across coupling hubs, parallel <= 0.05, angular <= 0.1 deg
Springs: 2 identical (wire ~1.2, OD ~10, free ~30); rate calibrated with 2 kg and 5 kg masses
Studs M5 × 110 (8.8), base lock-nut; nut + lock-nut above washer; never run with nuts removed
Guard closed + supply OFF/E-stop pressed for every adjustment (isolation rule)

IVHM bearing rig - Design B - machined parts
Rev A, 22 Oct 2026 - sheet 2/2 - all dims mm
General tol. ISO 2768-m; (V) = verify on purchased part before machining